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DIGIPLEX

DIGIPLEX is a technique which allows digital test, using the 4 digital channels of the ACLAM by multiplexing them through the scanner or, in the Flying Prober through the SCAFP and the probes. (DIGItal multiPLEXer)

Features

- Use the VL digital language and the ICT digital pattern generation.
- All the analog channels (scanner) can be used as normal digital channel.
- Possibility to use more than 4 resources (by grouping channels of the same logic state).
- Controlled Edge generation (non-switched, the commutation is made using the Digital Channels instead of the Relays, it is to avoid the oscillation of the status).
- Auto-correction (pin not connected, pin linked together and/or grounded).
- Use the various kind of channels in mixed mode (F50, MIS32,VBN, DIGIPLEX, FLY).
- It may also use the virtual flying channel in any combination (MAX 4-8 channels).
- It is possible to insert a user analog test (1 line only) in mixed mode with digital test, for example in the case of a test on a DAC or ADC Converters
- It is possible to automatically insert a pull-up/pull-down resistor on the output. (test of the OC).
- It is possible to insert a load programmable in voltage and current ±100V 0-500ma (Test fan-out). It is to check the max number of logic port that is possible to connect to the Digital output to avoid over-current errors.

Limitations

- It can't be used in dynamic mode
- Current limit of 400ma max of back-driving (in case of pin grouping the current needed may exceed the limit)

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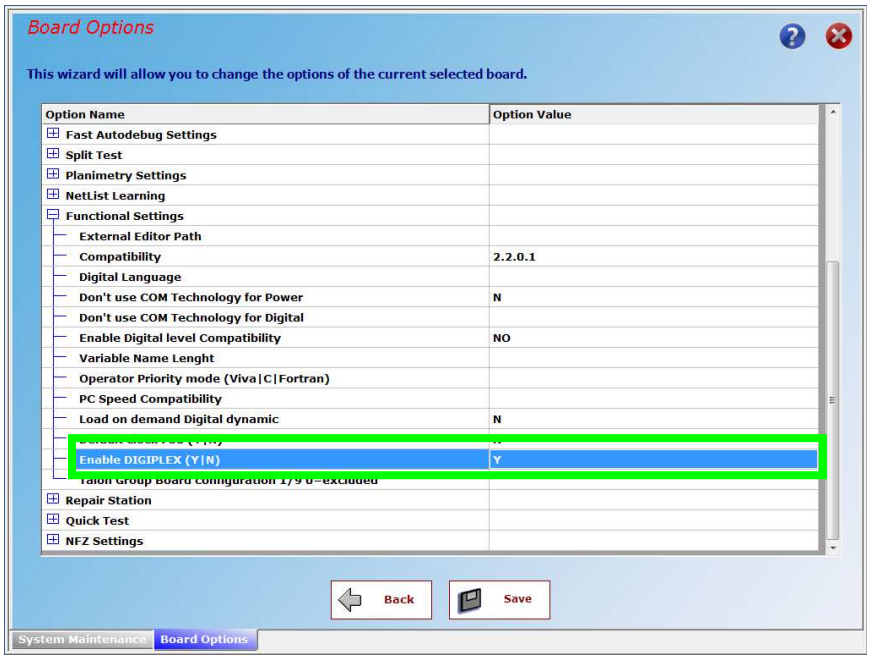
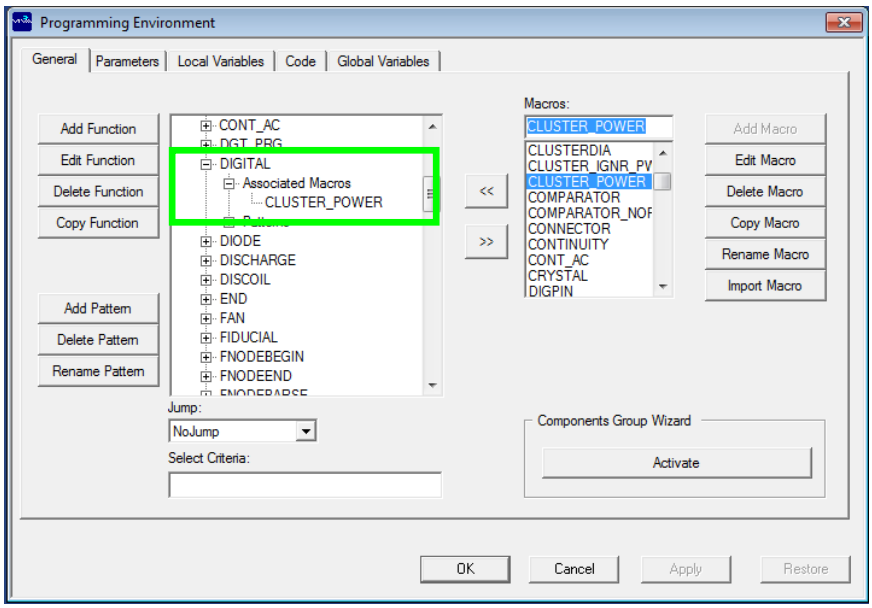
How to Enable and Assign the DIGIPLEX macro

- To enable the Digiplex set to **Y** the value in the *Functional Settings* in Board/Template Option



- In the Programming Environment check the associated macro to the function DIGITAL. It must be CLUSTER_POWER



If the associated macro is wrong (on Release before 3.0 the associated macro was CLUSTER), select the macro and with the button >> remove it from the associated macro list. Then search and select the macro CLUSTER_POWER from the macros list and add it with the button <<. Then apply, save and exit.

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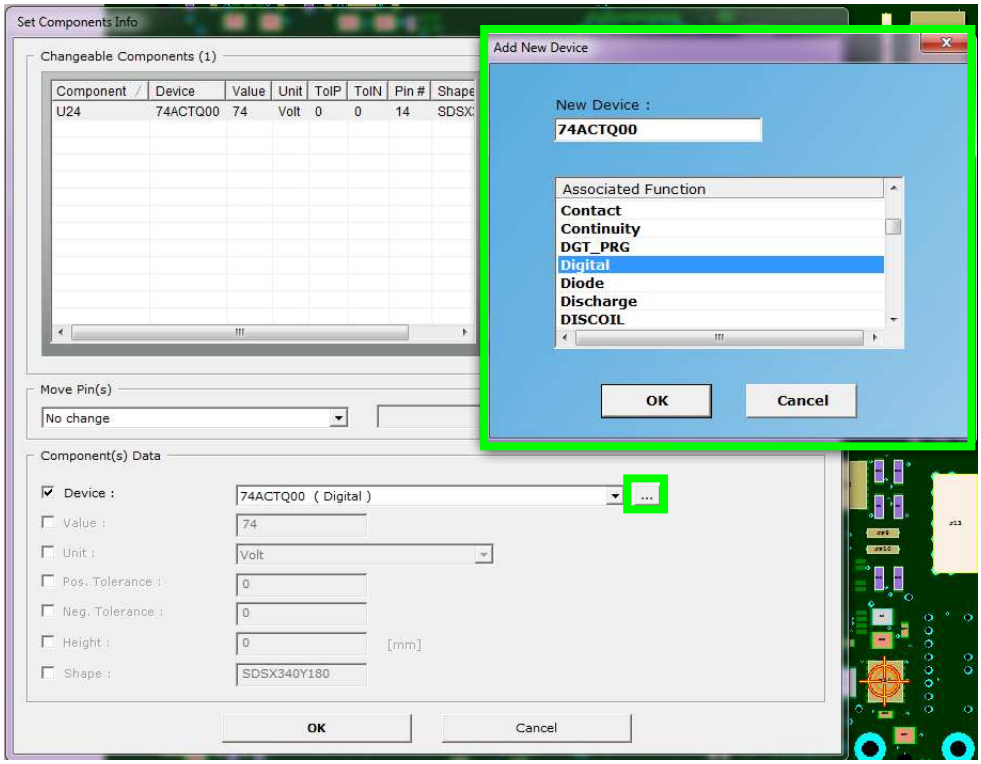
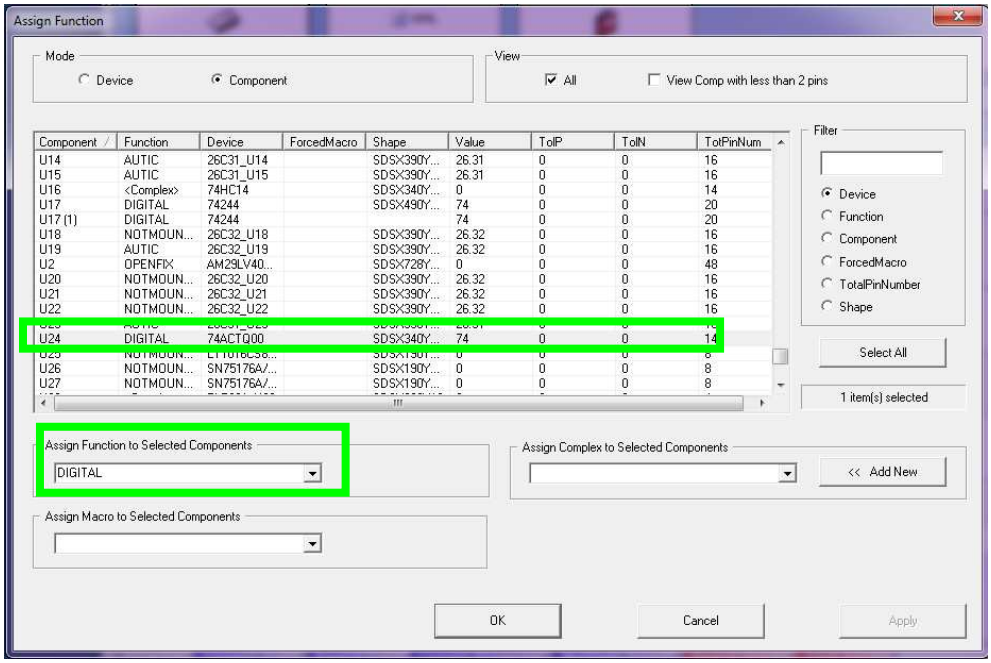
- Assign the device to the component under test if not already done during BoM import (e.g.: U24). The device 74ACTQ00 is included in the list of the Alias for the device 7400 and then the correct instructions will be associated to the component (more details about libraries and .PAT file will be explained in the next paragraphs)



- Assign the digital function to the component if not already done during BoM import.



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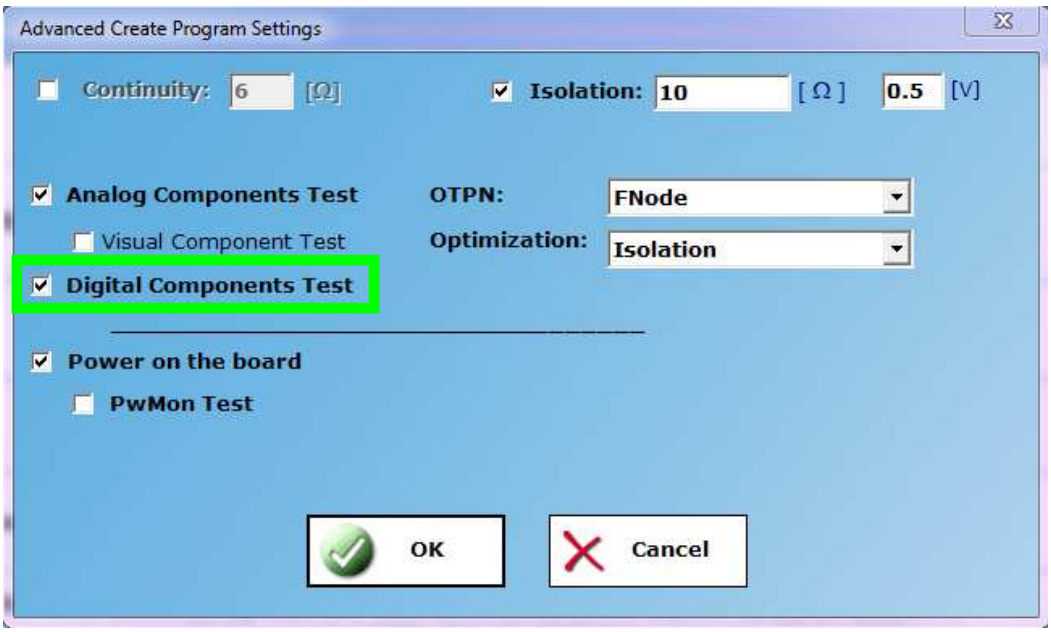
- Tick the option *Digital Component Test* in the Advanced Create Program Settings before to generate or re-generate the program test



HOW DOES IT WORK

- Whenever during execution VIVA encounters a digital command for an analog channel, the software automatically connects one of the ACLAM Digital Channels to the analog channel, and sets the specific digital state. In order to optimize the number of relay switching, VIVA uses the digital channels in random mode.
- The four digital channels are used according to the following scheme:
  1- Connection of one or more pins of the UUT to logic level L
  2- Connection of one or more pins of the UUT to logic level H
  3- Use of the channel for the generation of a single signal, i.e. the CLOCK
  4- Use of the channel to act as a SENSOR, one output at a time, repeating the test.

The architecture allows generation of both combinatorial or sequential truth-tables in a simple but effective way. The only possible limitation is if overdrive current is exceeded when many pins need to be forced to logic 1 or logic 0 simultaneously.

When one of the four functions is not needed, the software will automatically use the free channel to drive signals in single mode.

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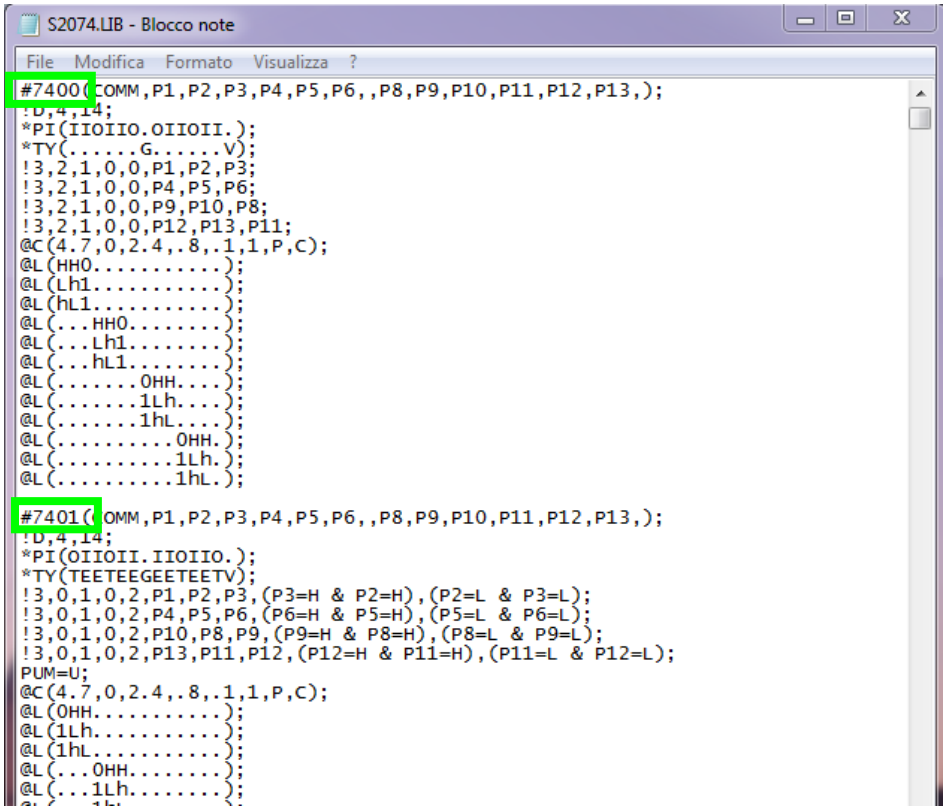
LIBRARY and PATTERN structure

The **.PAT** file is automatically created for all components with function DIGITAL assigned, and it has written inside all the instructions for the test of the component. Depending on the DEVICE NAME, Viva searches in the *libraries* (**.LIB**) the availability of a truth-table and creates the corresponding cluster-test with the associated pinlist. If there's no matches in the libraries, the file with the pinlist is crated anyway and it is to be filled-in by the user. All the libraries are stored in the VivaBoard folder. Below the list of the default libraries available:

- S2074.LIB
- S2010.LIB
- S20MIX.LIB
- S20MT.LIB

Each library has written inside all the information to test a family of components. Like sample the library **S2074.LIB** will be analyzed below.

To identify the family the library name must be divided: "**S20**" is the acronym of the system and "**74**" is the family of components. In the file are listed the different family types, like 7400, 7410, 7402, ..., and each type has its defined configuration and truth-table.



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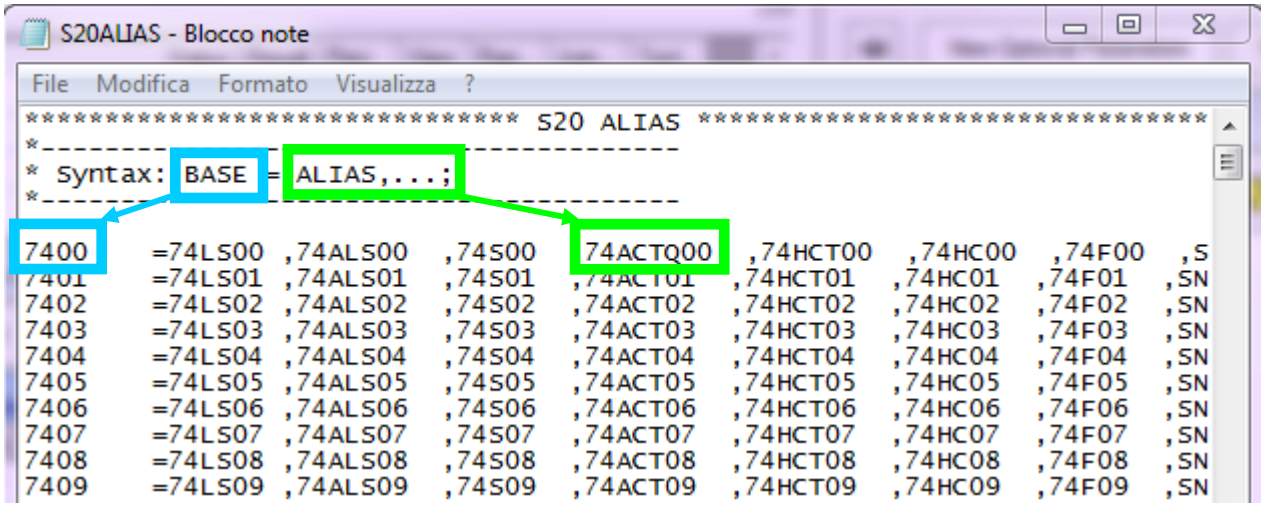
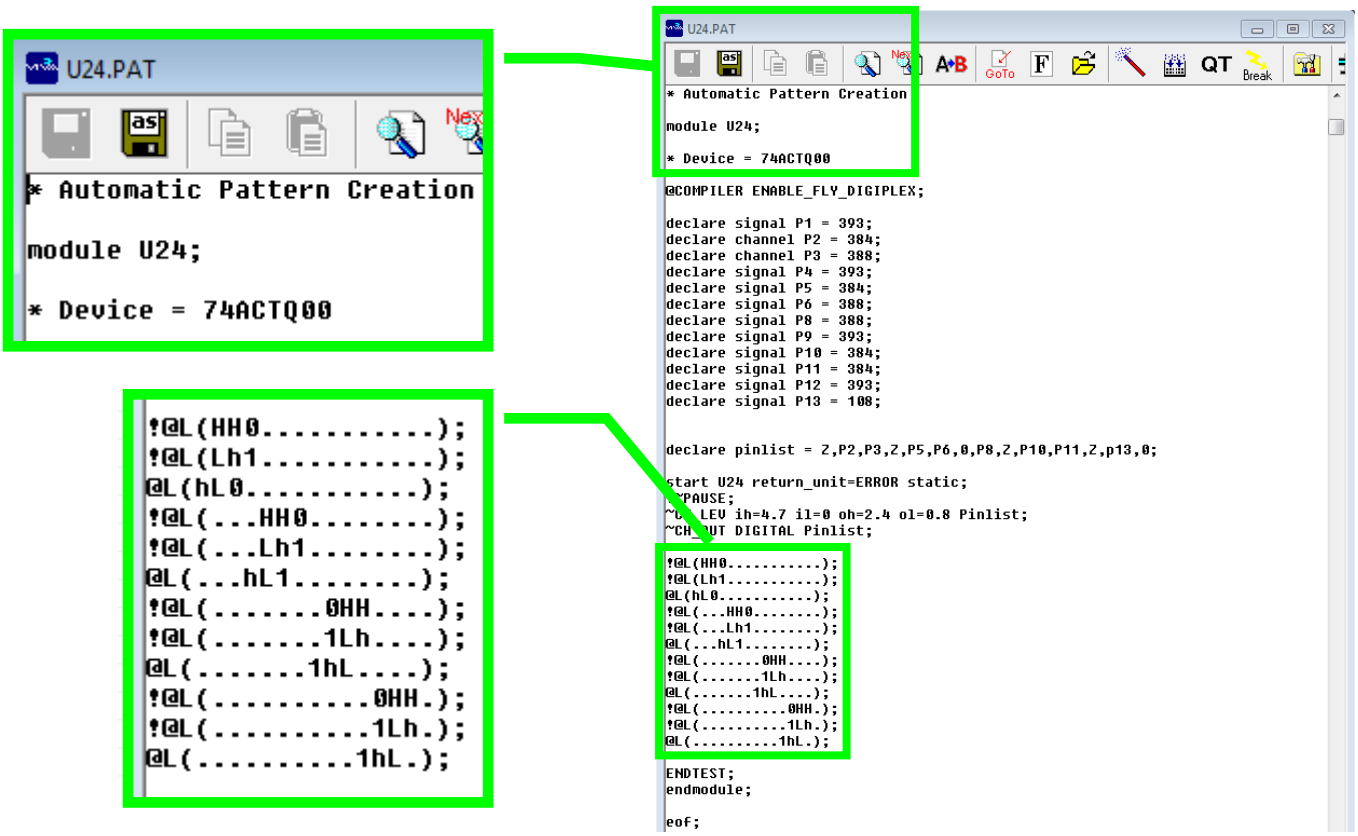
The correct configuration and truth-table will be copied into the .PAT file of the component. Of course the DEVICE of the components will not be the name of the family type (like 7400) but it will be for example *74ACTQ00*. In the VivaBoard folder is stored the file

- S20ALIAS

In this file are written all the alias to identify the correct family type.



Below a picture of the .PAT file of the component U24 created from the file S2074.LIB.



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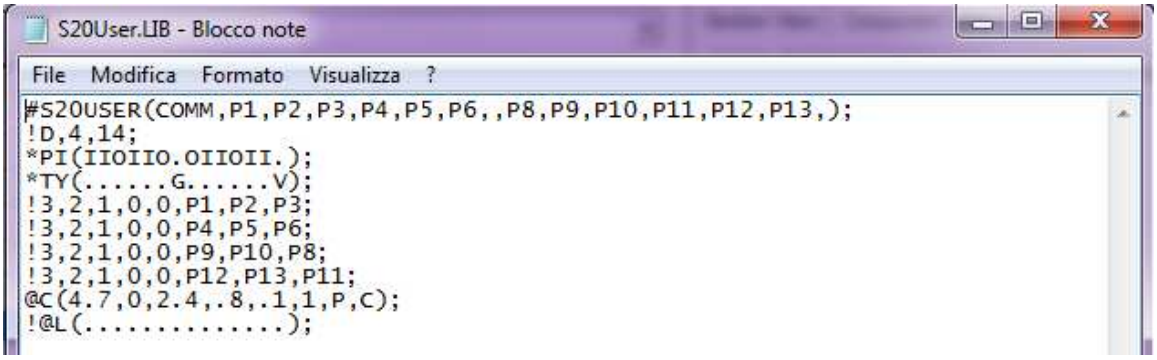
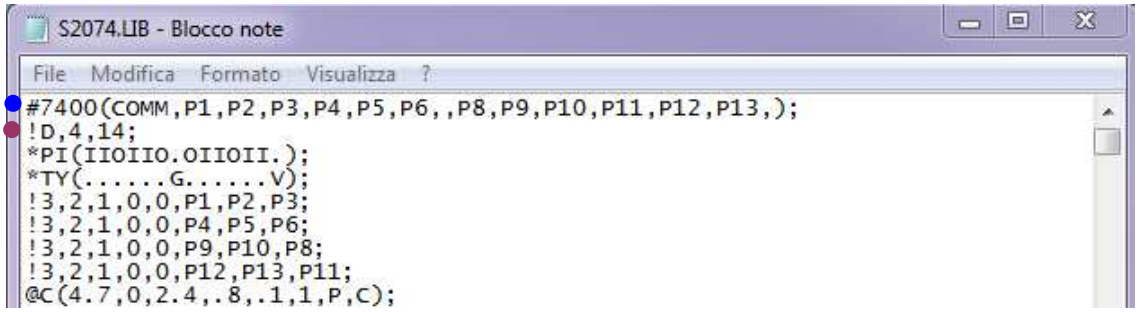
The last standard library stored in the VivaBoard folder is the file

- S20User.LIB

This file contains the general information about a digital component but not the information about the truth-table. If Viva can't find the device in the library (Alias included), the .PAT file will be created starting from the information written in the S20User.LIB file also the pinlist will be written but the truth-table will be empty.



SYNTAX and SYMBOLS



#CompName(Comm,Pin1,Pin2,....);

CompName	→	String (max 20 char.) with the component name
Comm	→	Comment
Pin1,Pin2,....	→	List of the pins, included VCC and GND (pins not declared in the list, e.g.: P7 and P14)

!CompType,SectionNr,CompPinNr;

CompType	→	Type of the component A : Analog D : Digital X : Not Defined

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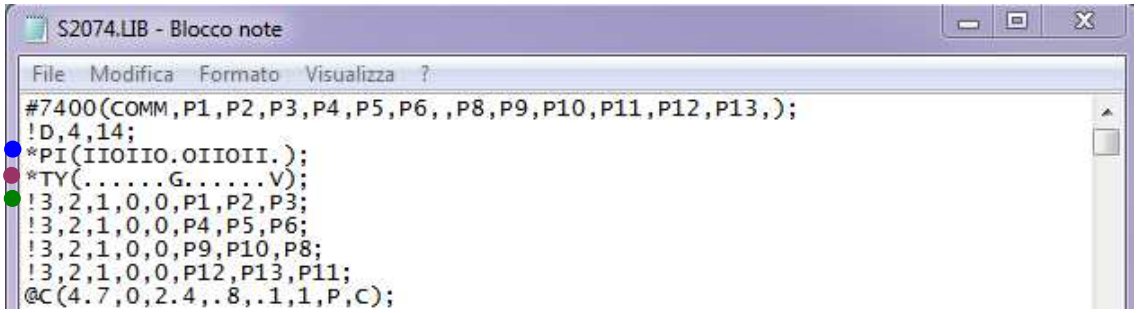

***PI(PinInformation);**

PinInformation → Declare the pins as Input (**I**) or Output (**O**)

***TY(PinType);**

PinType → Declare the analog pin type GND (**G**) and VCC (**V**)

!SectionPinNr,InPinNr,OutPinNr,BidirPinNr,EnablePinNr,Pin1,…;

SectionPinNr → Total number of the pins in the section
InPinNr → Number of the Input pins in the section
OutPinNr → Number of the Output pins in the section
BidirPinNr → Number of the Bidirectional pins in the section
EnablePinNr → Number of the Enable pins in the section
Pin1,… → Pins name

Each component typically has more than one section, this means that in the file will be more than one line with the instruction above. Below a sample of two section line.

!3,2,1,0,0,P1,P2,P3 Section 1
→ 3 total pins in the section
→ 2 Input pin
→ 1 output pin
→ 0 bidirectional pin
→ 0 enable pin
→ P1,P2 are the Input pins
→ P3 is the Output pin

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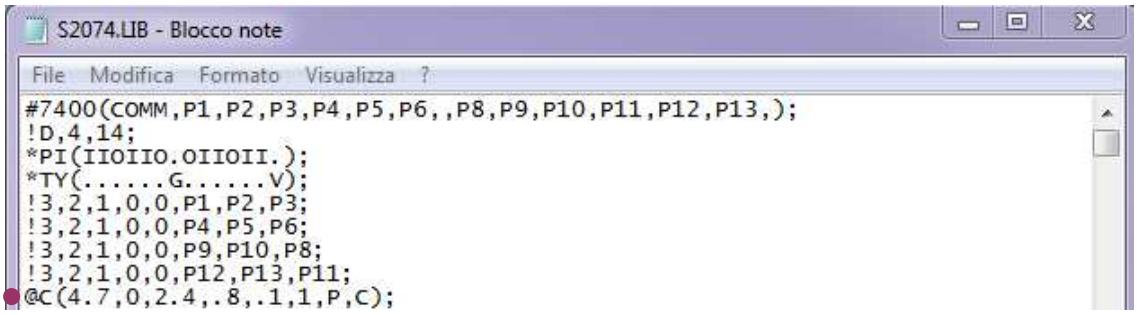
!3,2,1,0,0,P4,P5,P6 Section 2

→ 3 total pins in the section
→ 2 Input pin
→ 1 output pin
→ 0 bidirectional pin
→ 0 enable pin
→ P4,P5 are the Input pins
→ P6 is the Output pin

In case of enable pins present, the form it's the following

!3,1,1,0,1,P1,P2,P3,(P3=L),(P3=H); Section

→ 3 total pins in the section
→ 1 Input pin
→ 1 output pin
→ 0 bidirectional pin
→ 1 enable pin
→ P1 is the Input pins
→ P2 is the Output pin
→ P3 is the enable pin
→ (P3=L),(P3=H) means that the section is enable if the logic level of P3 is LOW and the section is disable if the logic level of P3 is HIGH



@C(InHVal,InLVal,OutHVal,OutLVal,...);

InHVal	→	Input voltage value for High logic level
InLVal	→	Input voltage value for Low logic level
OutHVal	→	Output voltage value for High logic level
OutLVal	→	Output voltage value for Low logic level

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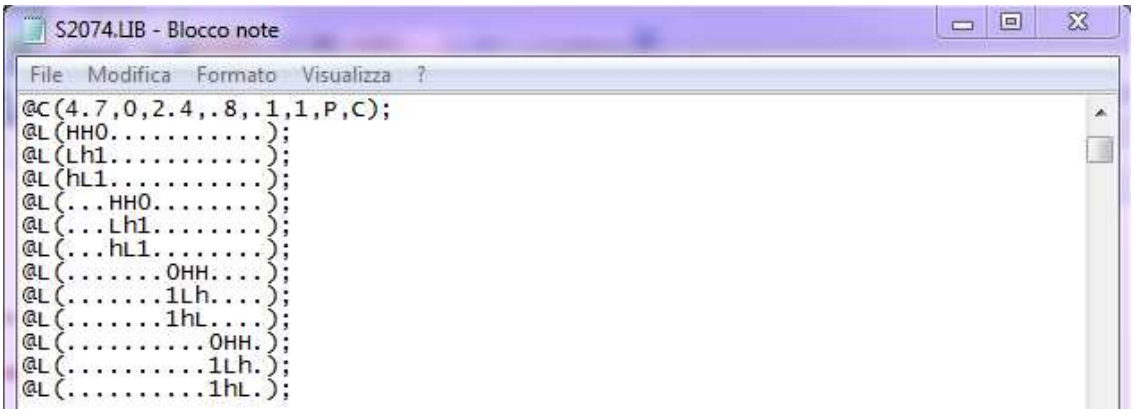

@L( );
.
.
@L( );

Each line who starts with **@L** identifies a truth-table line. In the sample, shown in the picture above, there are 12 lines (3 lines for each section of the component) and in each line are written the instructions to force the logic level of the input pins and to read the logic level of the outputs.

To write the commands to force and read the logic level on the component, the following symbols must be used. It is necessary to insert one symbol for each pin in each section involved in the single test (@L line).

H	→	Input logic level forced to HIGH
h	→	Input logic level forced to HIGH, IGNORED in debug or in case of error during the function generation
L	→	Input logic level forced to LOW
l	→	Input logic level forced to LOW, IGNORED in debug or in case of error during the function generation
T	→	CLOCK
.	→	HIGH IMPEDANCE
1	→	Output HIGH logic level
0	→	Output LOW logic level

.

During compilation VIVA generates an error if, due to the constraints, it is not possible to generate the digital functions for the symbols HL01T (while ignored for the symbols hl). For example the instruction

@L(LH0.....);

generates error if pins 1 and 2 are connected together.

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Instead the instruction

@L(Lh0.....);

doesn't generate error even if the pins 1 and 2 are connected together.

In addition to the truth-table, it is possible to add other instructions into the library file or directly into the .PAT file to improve the component test. The following syntax must be used:

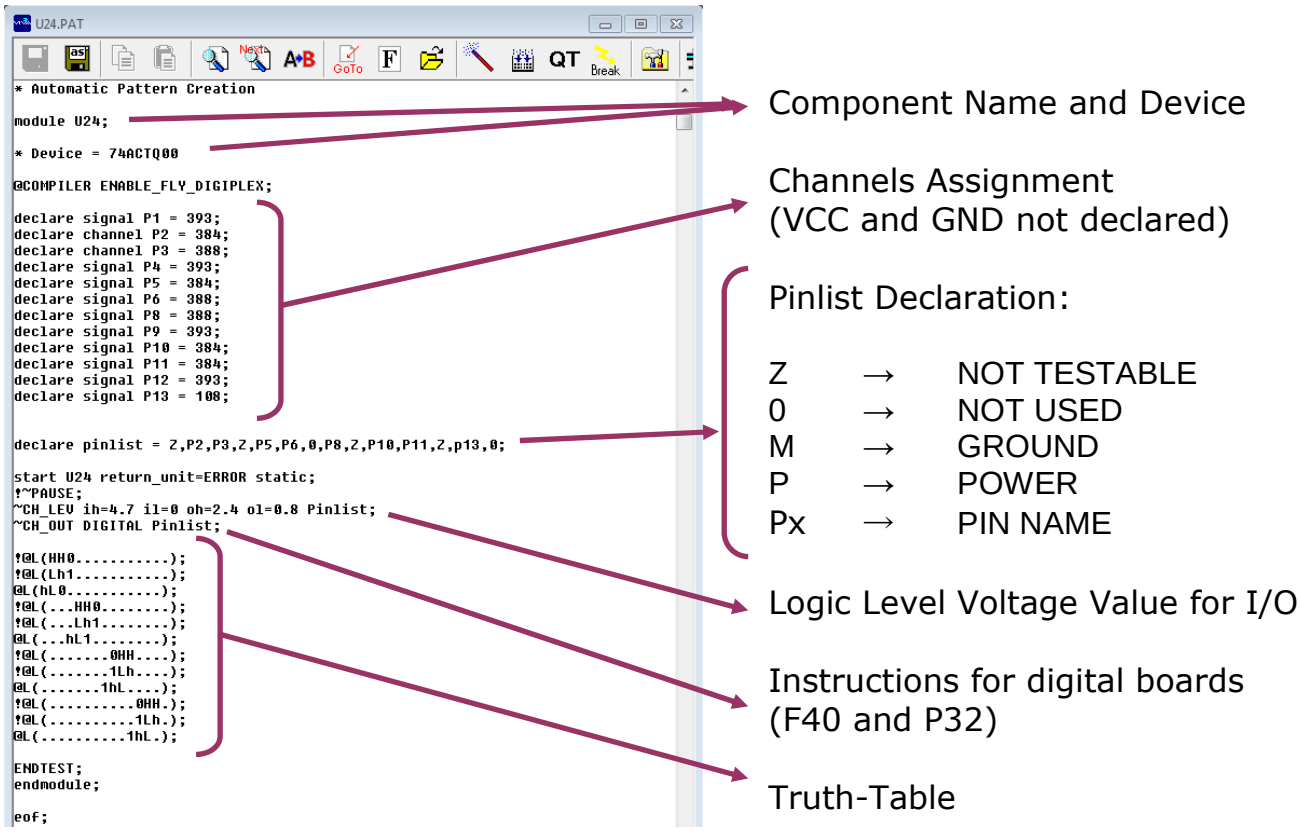
@PAT;

… instructions …

@PAT;

AUTOMATIC PATTERN CREATION

Starting from the information written in the .LIB file, the Viva SW will automatically create the .PAT file for each component declared as DIGITAL. In the .PAT file will be written the information about the channels assignment and the truth-table (if the device of the component is found in the libraries or in the alias files); the channels/signals assignment will be done anyway.



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\\ THIS SECTION IS NOT COMPLETE \\

FUNCIONAL DEBUG ENVIRONMENT

Descrizione aperture vari ambienti di test e debug con il debug on error abilitato e mostrare le varie videate con errori etc…

ADVANCED TESTS

- TEST OF THE OPEN COLLECTOR.

  Add the command ~SET DIGIPLEX PULL UP 1K; (forzare uscita alta)

- TEST OF THE OPEN DRAIN
  Add the command ~SET DIGIPLEX PULL DOWN 100; (forzare uscita bassa)

- TEST WHIT LOAD
  Add the command ~SET DIGIPLEX LOAD V=4.5 I=0.2;
  Load are programmable from 0 to 100v 0-100ma or 0-10v 0-500ma (Fan Out test per verifica carichi)

- TEST MIXED MODE DEVICE.
  Add the command ~SET DIGIPLEX HOLD L3;
  The selected line will be kept free by the DIGIPLEX ™. (per canale analogico)